Here to Slay DUNGEONS — Full Codebase Audit

Generated 2026-04-02 — Covers index.html, app.js (3702 lines), styles.css (1205 lines), and all assets

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A. High-Level App Summary

This is a **single-page, vanilla JS web app** for planning dungeon tile layouts in the *Here to Slay: DUNGEONS* board game expansion. It lets you:

1. Pick a dungeon tile set (Molten, Overgrown, etc.)
2. Randomly draw 6 of 9 available tiles plus the mandatory entrance tile
3. Drag tiles onto a hex-snapped board with rotation and contact-rule validation
4. Mark wall faces on tiles (blocking adjacency)
5. Place boss tokens on the board
6. Swap reserve tiles into the active tray
7. Save/load dungeon layouts as JSON

No framework, no build step, no bundler, no TypeScript. One HTML file, one JS file (3702 lines), one CSS file (1205 lines). Everything is imperative DOM manipulation.

B. File Tree with Roles

```
Here_to_slay/
├── index.html              Entry point. Static HTML shell.
├── app.js                  ALL logic: state, rendering, drag/drop, validation,
│                           persistence, wall editor, boss tokens, hex grid. 3702 lines.
├── styles.css              ALL styling. 1205 lines. Two themes (default + molten dark).
├── hex-background.html     DEAD FILE. Standalone hex demo absorbed into app.js.
├── icon.png                App icon (663KB — oversized)
├── icons/
│   ├── hand.png            DEAD FILE. Not referenced anywhere.
│   ├── reroll.png          Tray "Reroll Tiles" icon
│   └── reset.png           Tray "Reset Tiles" icon
├── gfx/                    Promotional/reference images (not used by app)
├── tiles/
│   ├── molten/             ONLY COMPLETE SET: entrance, 1-9, 2 bosses, reference
│   ├── overgrown/          PARTIAL: 2 bosses + reference only, NO numbered tiles
│   ├── dreamscape/         EMPTY
│   ├── nightmare/          EMPTY
│   ├── submerged/          EMPTY
│   └── deep_freeze/        EMPTY
├── slay info/
│   └── HTS_Dungeons_Rules…pdf    Rules reference
├── CHANGELOG.md
├── README.md
└── .gitignore
```

Issues Found

File	Problem
`app.js`	**MONOLITH** — 3702 lines handling everything. #1 structural problem.
`hex-background.html`	**DEAD** — Logic absorbed into `app.js` `renderBoardHexGrid()`.
`icons/hand.png`	**DEAD** — Not referenced in HTML or JS.
`tiles/overgrown/overgrown_boss_rootgnaw`	**BUG** — Missing `.png` extension on disk AND in `BOSS_TILE_FILES` (app.js:29).
`app.js:153` (and ~15 other places)	**BUG** — Entrance tile ID hardcoded as `"molten_entrance"` regardless of selected tile set.
`icon.png`	663KB for an app icon — should be optimized.
4 of 6 tile set folders	Completely empty. Dropdown offers all 6 but only Molten works.

C. Main UI Map

The app has **two views**, toggled by `body.wall-edit-mode`:

View 1: Build Dungeon (default)

```
┌──────────────────────────────────────────────────────────────┐
│ TOPBAR (.topbar)                                             │
│  Title + subtitle │ [UI Theme] [Debug▾] [Settings▾]          │
│                    [Edit Walls] [wall-edit-only buttons…]    │
├──────────────────────────────────────┬──────────────────────┤
│ BOARD (.board)                       │ SIDEBAR (.sidebar)   │
│  ┌──────────────────────┐            │  TRAY HEAD           │
│  │ .board-hex-grid (SVG)│            │   "Tile Tray" [🔄][↺] │
│  │ .board-content       │            │   [Tile Set ▼]       │
│  │   tiles, reference,  │            │  TRAY (#tray)        │
│  │   boss tokens        │            │   2-col × .tray-slot │
│  │                      │            │  RESERVE+BOSS ROW    │
│  │   [Zoom XX%]         │            │   [Reserve] [Boss]   │
│  └──────────────────────┘            │  STATUS BAR (.status)│
│  DRAG LAYER (.drag-layer)            │                      │
└──────────────────────────────────────┴──────────────────────┘
```

View 2: Wall Editor

```
┌──────────────────────────────────────────────────────────────┐
│ TOPBAR (+ [Frontpage] [Clear Walls] [Export] [Import])       │
├──────────────────────────────────────────────────────────────┤
│ WALL EDITOR PAGE (#wall-editor-page)                         │
│  Horizontally scrolling theme panels with clickable tiles    │
└──────────────────────────────────────────────────────────────┘
```

Component-to-Code Mapping

Visible Region	Rendered By	Container
Dungeon set selector	`<select id="theme-select">`	`.tray-pickers`
UI theme selector	`<select id="ui-theme-select">`	`.actions`

Debug toggles	`<details class="view-menu">`	`.actions`
Settings menu	`<details class="settings-menu">`	`.actions`
Tile tray (6 slots)	`renderActiveTiles()` / `rerenderTrayAndReserve()`	`#tray`
Reserve pile (3 inactive)	`renderInactiveTileStack()`	`#reserve-stack`
Boss stack	`renderBossStack()`	`#boss-stack`
Dungeon board	`placeStartTileAtCenter()`, drag/drop	`#board > .board-content`
Hex grid	`renderBoardHexGrid()`	`.board-hex-grid` (SVG)
Drag ghost	`beginDrag()` reparents to `.drag-layer`	`.drag-layer`
Boss token (on board)	`createBossToken()`	`.board-content`
Reference card	`placeReferenceAboveStart()`	`.board-content`
Zoom indicator	`updateBoardZoomIndicator()`	`#board`
Rotate controls	`createTileElement()`	`.tile-controls`
Status bar	`setStatus()`	`#status`
Save/Load	`exportDungeonLayout()` / `importDungeonLayout()`	Settings dropdown
Wall editor page	`renderWallEditorPage()`	`#wall-editor-page`

CSS Architecture

- **Design tokens** via `:root` variables: `--bg`, `--panel`, `--ink`, `--muted`, `--accent`, `--accent-soft`, `--warn`, `--tile-size`
- **Molten theme**: `body.ui-theme-molten` overrides (~80 lines of dark theme selectors)
- **Layout**: `.workspace` is `grid-template-columns: minmax(0,1fr) 390px`
- **Responsive**: single `@media (max-width: 980px)` breakpoint
- **State-driven body classes**: `wall-edit-mode`, `show-guide-labels`, `show-wall-faces`, `reserve-edit-mode`, `reserve-swap-pending`, `boss-edit-mode`, `feedback-legacy`, `ui-theme-molten`
- **z-index layers**: `.drag-layer` 20000, `.boss-token-preview` 26000, `.tile-controls` 12000, dropdowns 40000

D. State and Data-Flow Map

Central State Object (app.js:64-93)

```
const state = {
  tiles: Map<string, TileState>,     // tile id → full tile object
  tileDefs: TileDef[],               // current tile set definitions
  selectedTileId: string | null,
  hoveredTileId: string | null,

  selectedThemeId: string,           // "molten", "overgrown", etc. (tile set)
  selectedUiThemeId: string,         // "current" or "molten" (color scheme)

  showGuideLabels: boolean,          // debug: face index numbers
  showWallFaces: boolean,            // debug: wall highlights
  ignoreContactRule: boolean,        // bypass 4-point validation
```

```
  useFaceFeedback: boolean,          // face-by-face vs full-outline

  wallEditMode: boolean,
  wallOverrides: { [themeId]: { [tileId]: number[] } },
  wallEditorTileRefs: Map,
  wallEditorActiveThemeId: string | null,
  wallEditorActiveTileId: string | null,

  reserveEditMode: boolean,
  reserveOrder: string[],
  pendingSwapSource: { zone: "tray"|"reserve", tileId } | null,

  bossEditMode: boolean,
  bossOrderFlipped: boolean,
  bossTokens: BossToken[],
  nextBossTokenId: number,

  boardPanX: number, boardPanY: number, boardZoom: number,
  frontpageSnapshot: Snapshot | null,
  referenceTileSrc: string,
  referenceMarker: { dom, x, y } | null,
};
```

TileState Shape

```
{
  id, src, required,                // from TileDef
  img: HTMLImageElement,            // loaded image
  x, y: number,                    // position (board or tray coords)
  rotation: number,                 // 0, 60, 120, 180, 240, 300
  placed: boolean, active: boolean,
  dom, bodyDom, guideDom, traySlot, // DOM references
  drag: DragState | null,
  invalidReturnTimer: number | null,
  shape: { minX, minY, maxX, maxY, radius },
  alphaMask: { width, height, alpha: Uint8Array },
  faceGeometry: { points, normals, avgSideLength, avgOffset },
  sideLength, apothem: number,
  wallFaceSet: Set<number>,         // face indices marked as walls
}
```

Derived vs Stored

Data	Source	Where
Which 6 tiles are active	**Computed** per round	`startRound()` via `shuffle().slice(0,6)`
Tile face geometry	**Derived** from PNG pixel data	`getFaceGeometry()`, `getAlphaMask()`
Contact validity	**Derived** on every drop	`findBestContact()`
Wall face set	**Stored** in localStorage	Key: `hts_wall_overrides_v1`
UI theme preference	**Stored** in localStorage	Key: `hts_ui_theme_v1`
Dungeon layout	**Exported** as JSON file	`exportDungeonLayout()`
Boss token positions	**In-memory only** — NOT saved	Lost on refresh or save/load

Persistence Formats

Wall overrides (localStorage):

```
{ "molten": { "tile1": [3, 7, 12], "tile5": [0, 14] }, "overgrown": {} }
```

Dungeon layout (exported JSON file):

```
{
  "schema": "hts-dungeon-layout-v1",
  "savedAt": "2026-04-02T…",
  "layout": {
    "themeId": "molten",
    "boardPanX": 0, "boardPanY": 0, "boardZoom": 1,
    "reserveOrder": ["tile3", "tile7", "tile9"],
    "tiles": [
      { "id": "molten_entrance", "active": true, "placed": true,
        "x": 400, "y": 240, "rotation": 0 }
    ]
  }
}
```

Notable gap: Boss tokens and wall data are NOT included in dungeon layout saves.

E. Naming Glossary

Concept	Preferred Name	Current Code Name	Avoid	Rename?
The app	**Dungeon Builder**	"Tile Randomizer Prototype"	"randomizer"	Yes — update title
Tile set	**Tile Set**	`themeId`, `selectedThemeId`	"theme" alone	**Yes → `tileSetId`**
UI color scheme	**UI Theme**	`selectedUiThemeId`	"theme"	OK as-is
Entrance tile	**Entrance Tile**	`"molten_entrance"`	"start tile"	**BUG: use `"entrance"`**
Regular tiles	**Dungeon Tiles 1-9**	`"tile1"–"tile9"`	"cards"	—
Active tiles in tray	**Tray Tiles**	`tile.active && !tile.placed`	"selected"	—
Reserve pile	**Reserve Pile**	`#reserve-stack`	"stack"	—
Boss stack	**Boss Pile**	`#boss-stack`	—	—
Boss on board	**Boss Token**	`.boss-token`	"boss card"	—
Boss in pile	**Boss Card**	`.boss-card`	"boss tile"	—
Reference card	**Reference Card**	`.board-reference-tile`	"legend"	—
Wall face	**Wall / Wall Edge**	`wallFaceSet`	"blocked face"	—
Contact rule	**Contact Rule**	`MIN_CONTACT_POINTS = 4`	"adjacency"	—
Blocked A/B points	**Entrance Blocked Edges**	`TILE_START_NON_NUMERIC_POINTS`	"non-numeric"	**Yes — rename**
Debug menu	**Debug Menu**	`.view-menu`	"view menu"	**Yes — rename class**

Main view	**Build View**	"frontpage"	"home"	—
Reroll	**Reroll**	`rerollTrayTiles()`	"shuffle"	—
Reset	**Reset Tiles**	`resetTiles()`	"clear"	—
Swap	**Reserve Swap**	`performReserveToTraySwap()`	—	—

F. Rules-to-Code Mapping

1. "6 of 9" Tile Selection

- **Where:** `startRound()` at lines 693-694
- **How:** `shuffle(candidateTiles).slice(0, 6)`
- **Assessment:** Correct.

2. Entrance Tile Handling

- **Where:** Hardcoded as `"molten_entrance"` in ~15 locations
- **How:** Always active, auto-placed at center, rotation locked once other tiles placed
- **BUG:** ID is `"molten_entrance"` even for non-Molten sets.

3. Adjacency / Contact Validation (4-Point Rule)

- **Where:** `findBestContact()` L2675, `getContactMatchDetails()` L2708
- **How:** For each face pair, checks opposite normals (dot < -0.88), tangent alignment (> 0.85), midpoint distance. Uses longest-increasing-subsequence for ordered face matching. Total contact = matched faces × 2. Must reach ≥ 4 points.
- **Assessment:** Sophisticated and well-implemented. Pixel-geometry-based.

4. Wall Blocking

- **Where:** `getContactMatchDetails()` L2730: `if (af.isWall || bf.isWall) continue;`
- **How:** Wall faces excluded from contact counting.
- **Limitation:** Wall data is manually authored. No defaults ship with app.

5. Entrance Blocked Edges (A/B)

- **Where:** `TILE_START_NON_NUMERIC_POINTS = new Set([11, 12])`, `isBlockedContactFace()` L3034
- **How:** Points 11/12 of entrance are blocked. Opaque pixel touch test prevents placement against doorway.
- **Assessment:** Correct per game rules.

6. Rotation

- **Where:** `rotateTile()` L2648 — 60° increments, normalized to 0-360
- **How:** CSS rotation on `.tile-body`; math rotation in `getContactFaces()`
- **Assessment:** Correct.

7. Boss Filtering by Dungeon

- **Where:** `BOSS_TILE_FILES` constant L27-30
- **How:** Hardcoded map. Only Molten and Overgrown have entries.
- **BUG:** `"overgrown_boss_rootgnaw"` missing `.png` extension in code and on disk.

8. Save/Load

- **Where:** `exportDungeonLayout()` L626
- **Content:** Theme, pan/zoom, reserve order, per-tile state
- **Missing:** Boss token positions, wall data, UI theme.

Brittleness Concerns

#	Issue	Severity
1	Hardcoded `"molten_entrance"` ID blocks multi-set support	Critical
2	Guide point adjustments (~200 lines of pixel nudges) are fragile magic numbers	High
3	No default wall data — fresh install has no walls	High
4	No schema migration for save files or localStorage	Medium
5	No tile count enforcement (can place 1 or 6 with no feedback)	Low

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G. UX Problems

Critical

1. **Only one tile set works.** Selecting anything except Molten silently fails.
2. **"Theme" means two things.** Tile set and UI color scheme both use "theme" in code.
3. **Instruction text shows code identifiers.** Header says "Place `molten_entrance`".
4. **No default wall data.** Wall-blocking validation is effectively disabled on fresh install.

High Friction

5. **Save/Load buried in Settings dropdown.** Primary actions hidden behind secondary menu.
6. **Boss tokens vanish on save/load.** No indication they aren't saved.
7. **Debug menu mislabeled.** Useful toggles hidden under "Debug" label.
8. **Reserve/boss interaction invisible.** Click-to-toggle with no visible affordance.

Minor

9. **No undo.** Misplaced tiles have 10-second auto-return. No general undo.
10. **Status bar easily missed.** All feedback goes to small text at sidebar bottom.
11. **Keyboard shortcuts undocumented.** R/F for rotation, drag for pan, scroll for zoom.
12. **No tile count indicator.** No "3/6 placed" progress.

H. Refactor Recommendations

Do Now (Before Adding Features)

1. **Parameterize the entrance tile ID.** Replace ~15 occurrences of `"molten_entrance"` with a computed value. Unblocks all tile sets.
2. **Rename `TILE_START_NON_NUMERIC_POINTS`** → `ENTRANCE_BLOCKED_FACE_INDICES`.
3. **Fix `overgrown_boss_rootgnaw`** — add `.png` extension to file and code.
4. **Delete dead files** — `hex-background.html`, `icons/hand.png`.
5. **Disable unavailable tile sets** in the dropdown or mark "(coming soon)".
6. **Include boss tokens in dungeon save/load.**
7. **Bundle default wall data** as a constant.

Do Later (For Scaling)

8. **Split `app.js` into ES modules:** constants, state, tile-data, board, drag, validation, tray, boss, wall-editor, save-load, ui, geometry, main.
9. **Rename `themeId` → `tileSetId`** across the codebase.
10. **For auto-build:** Extract validation into a pure module with no DOM dependencies.
11. **For cross-set boss randomization:** Create unified boss registry.
12. **For minion mixing:** Would need a new data layer — not currently modeled.

I. Top 10 Safest Improvements First

#	Change	Effort	Risk	Impact
1	Fix `overgrown_boss_rootgnaw` missing `.png` extension	1 min	Zero	Fixes broken asset
2	Delete `hex-background.html` and `icons/hand.png`	1 min	Zero	Removes dead files
3	Replace `molten_entrance` in `buildTileDefs()` with `${prefix}_entrance`	5 min	Low	Unblocks all tile sets
4	Update topbar instruction text to not show code identifiers	2 min	Zero	Better first impression
5	Disable unavailable tile sets in dropdown	15 min	Low	Prevents confusing errors
6	Add boss tokens to dungeon save/load payload	20 min	Low	Prevents data loss
7	Rename `.view-menu` to `.debug-menu` in CSS/HTML	5 min	Zero	Code matches UI label
8	Add tile count badge to reserve pile	10 min	Low	Better scannability
9	Move Save/Load to visible topbar buttons	10 min	Low	Core feature discoverable
10	Rename `themeId` → `tileSetId` across codebase	30 min	Medium	Eliminates naming confusion